Canvas Transfer App

Simple Project Specification

PROJECT GOAL	A focused Apple-platform tool for transferring and comparing drawings on physical canvas.

Project Goal

Build a simple cross-platform Apple app that helps artists transfer a drawing from paper to canvas using photographs, image alignment, layers, transparency, blend modes, and grids.

The app should work on **iPhone, iPad, and Mac**. The main use case is to photograph a reference drawing, photograph a canvas, align the two images, compare them while drawing, and repeat the process as the physical drawing develops.

Core Workflow

1. Create a new project and enter the physical canvas dimensions.
2. Take or import a photo of the source drawing.
3. Straighten the image, then resize, rotate, skew, and position it as needed.
4. Add an adjustable grid overlay.
5. Take or import a photo of the physical canvas and straighten/alignment it.
6. Reorder the source and canvas layers; apply Multiply and adjust opacity.
7. Draw physically on the canvas, photograph the updated canvas, and add it as a new layer.
8. Align and compare the new layer with the source drawing. Repeat as needed.
9. Save the project with all layers and transformations intact.

Required Features

Projects

- Create and name a project
- Enter canvas width and height
- Choose inches or centimeters
- Save and open projects
- Autosave changes

Image Input

- Take a photo using iPhone or iPad camera

- Import an image from Photos
- Import an image from Files

Image Straightening

Provide four draggable corner points. The user positions each point on a corner of the photographed drawing or canvas. The app corrects the perspective so the photographed surface becomes rectangular. Automatic rectangle detection can suggest the initial corners.

Image Transform

- Move
- Proportional scale
- Optional independent horizontal and vertical scale
- Rotate
- Horizontal skew
- Vertical skew
- Keep all transformations editable and nondestructive

Layers

- Add, delete, and rename layers
- Show or hide layers
- Reorder layers
- Select an active layer

Layer Appearance

- Opacity from 0-100%
- Normal blend mode
- Multiply blend mode

Grid

Add an editable grid layer. Controls should include horizontal divisions, vertical divisions, opacity, and line thickness. Common examples include 3 x 3, 4 x 5, and 8 x 10. The grid should match the proportions of the physical canvas.

Compare

- Overlay two layers
- Blink between two images
- Split view

Main Screens

Screen	Purpose
Project Browser	Shows saved projects and creates new projects.
New Project	Project name, canvas width, height, and units.
Workspace	Canvas view, layers, transforms, grid, compare, and Add Image.
Perspective Correction	Photo with four draggable corners plus Reset, Cancel, and Apply.

Suggested Technology

Component	Technology
Platform	Native Apple application
Programming language	Swift
User interface	SwiftUI
Image processing	Core Image
Camera	AVFoundation
Photo import	PhotosUI / PhotosPicker
Rectangle detection	Vision framework
Project files	Custom document package with images + project metadata

Core Image should handle transformations, perspective correction, compositing, opacity, and Multiply blending. Projects should remain nondestructive so original imported images are never permanently modified.

Basic Project Data

- **Project:** name, canvas width, canvas height, units
- **Layer:** name, image, visibility, opacity, blend mode, position, scale, rotation, skew, perspective corners
- **Grid:** rows, columns, opacity, line thickness

MVP

- Project creation and physical canvas dimensions
- Camera capture, Photo import, and File import
- Four-point perspective correction
- Move, scale, rotate, and skew

- Layer creation, reorder, rename, visibility, and deletion
- Opacity and Multiply blending
- Adjustable grid
- Overlay and Blink comparison
- Project saving and autosave
- iPhone, iPad, and Mac support

Future Features

- Automatic alignment between successive canvas photos
- Difference blend comparison
- Apple Pencil drawing
- Measurement tools and grid presets
- Automatic grid sizing from physical dimensions
- Color correction, crop tools, and masks
- Cloud collaboration and flattened comparison export

Success Criteria

The MVP succeeds when an artist can photograph a drawing, correct its perspective, add a grid, photograph and align a canvas, overlay the source using Multiply and transparency, make physical charcoal corrections, photograph successive versions, compare those versions with the reference, and later reopen the saved project with all layers and settings preserved.

Primary design goal: keep this workflow fast and simple rather than turning the application into a general-purpose image editor.